Critical Sibuya quantum resetting converges on every fixed finite unitary register

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We consider exact discrete-time unitary dynamics interrupted by unconditional resets to a fixed pure state. The reset hazard after n consecutive unitary steps is $r_n = \gamma/(n+1)$, with $0 < \gamma < 1$, so the inter-reset time has infinite mean. For every fixed finite-dimensional unitary, we prove that the ensemble density operator converges in Schatten trace norm to the pinching of the reset state onto the distinct eigenspaces of the unitary. The proof gives a nonasymptotic bound controlled by the occupied eigenphase chord gaps and by the two endpoint masses of the critical Sibuya age distribution. At a fixed nonzero gap, each coherence has two algebraic contributions, proportional to $t^{\gamma-1}$ and $z^t t^{-\gamma}$, from distinct unit-circle singularities. For $\gamma = 1/2$ and eigenphase ratio $z = -1$, every odd-time coherence vanishes exactly, whereas the even-time coefficient is a central binomial coefficient. The estimates concern the ordinary time sequence, not only an Abel limit or an adjacent-time diagnostic, and are intentionally nonuniform when an occupied eigenphase gap closes.

I. INTRODUCTION

Stochastic resetting replaces part of a system's history by repeated returns to a prescribed state. In a quantum register, unitary gates between reset events make the resulting ensemble a renewal mixture of coherent orbits. For a constant reset probability, the geometric age distribution gives a stationary state directly. Age-dependent reset probabilities are subtler: the age process is non-Markovian after the quantum state alone is retained, and an infinite-mean renewal law can keep appreciable weight at both ends of the available age interval.

Wald *et al.* formulated a discrete-time quantum reset model and studied both constant and age-dependent hazards [1]. For the power-law hazard $r_n = \gamma/(n+1)^\alpha$, their survival analysis and two-qubit adjacent-time numerics indicate relaxation at the critical exponent $\alpha = 1$. A vanishing adjacent difference does not, by itself, establish convergence of the ordinary state sequence. Independently, Michelitsch *et al.* derived the Sibuya backward-age distribution and its double generating function for discrete resetting on networks [2]. Their displayed relaxation formula treats transform arguments strictly inside the unit disk; nontrivial unitary eigenphases lie on its boundary. A related continuous-time quantum renewal model reaches a diagonal ensemble under a final-value assumption [3], but it does not provide the discrete critical ordinary-time bound considered here.

This paper resolves the critical finite-register question. The key elementary observation is that, at fixed clock time, the exact Sibuya age mass is U-shaped. Its total variation is therefore bounded by its two endpoint masses. Discrete summation by parts turns that fact into a Fourier bound at every nontrivial unit-modulus argument. A spectral-block decomposition then lifts the scalar estimate to Schatten trace norm. We also retain both boundary singularities of the generating function, which identifies the two competing decay powers and an exact parity cancellation.

All statements below have fixed finite Hilbert-space dimension, fixed unitary U, exact integer clock time, and $0 < \gamma < 1$. There is no continuum or perturbative approximation. We make no uniform claim as an occupied phase gap tends to zero, and no claim about other power-law exponents, infinite dimension, partial or conditional resets, dissipative interval evolution, or an experimental realization.

II. MODEL AND CRITICAL AGE LAW

Let $\mathcal{H} = \mathbb{C}^d$, with fixed finite d, and let $U \in U(d)$ be the unitary applied on a nonreset step. Fix a normalized reset vector $|\psi_R\rangle$ and $\rho_R = |\psi_R\rangle\langle\psi_R|$. The initial quantum state is ρ_R, and the initial age is zero. At age n, the next clock step resets the whole register to ρ_R with probability

$$r_n = \frac{\gamma}{n+1}, \qquad 0 < \gamma < 1, \tag{1}$$

or applies U with probability $1 - r_n$. A reset consumes one clock step, applies no additional gate, and returns the age to zero.

Writing $P_n(t)$ for the probability of age n at time t, the exact update and boundary data are

$$P_0(t+1) = \sum_{n=0}^{t} r_n P_n(t), \tag{2a}$$

$$P_{n+1}(t+1) = (1 - r_n) P_n(t), \tag{2b}$$

$$P_0(0) = 1, \qquad P_n(t) = 0 \quad (n < 0 \text{ or } n > t). \tag{2c}$$

The ensemble state is therefore

$$\rho(t) = \sum_{n=0}^{t} P_n(t) U^n \rho_R (U^\dagger)^n. \tag{3}$$

Every term is a unitary image of a normalized density operator and the weights below sum to one, so Eq. (3) is positive, Hermitian, and trace one at every finite time.

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For $k \geq 0$, define

$$a_k = \frac{(\gamma)_k}{k!} = \frac{\Gamma(k+\gamma)}{\Gamma(\gamma)\Gamma(k+1)},$$
$$b_k = \frac{(1-\gamma)_k}{k!} = \frac{\Gamma(k+1-\gamma)}{\Gamma(1-\gamma)\Gamma(k+1)}. \tag{4}$$

The survival probability of one excursion through age k is b_k, while a_k is the renewal mass of the Sibuya waiting law [2]. The last-renewal decomposition, equivalently induction in Eq. (2), gives the exact age law

$$\boxed{P_n(t) = a_{t-n} b_n, \qquad 0 \leq n \leq t.} \tag{5}$$

Normalization follows without an asymptotic approximation from the Vandermonde coefficient identity

$$\sum_{n=0}^{t} a_{t-n} b_n = [u^t](1-u)^{-\gamma}(1-u)^{\gamma-1} = 1. \tag{6}$$

The scalar factor multiplying a spectral coherence is the finite-age transform

$$F_t(z) = \sum_{n=0}^{t} P_n(t) z^n. \tag{7}$$

Its ordinary-time generating function is

$$\sum_{t \geq 0} F_t(z) u^t = (1-u)^{-\gamma}(1-zu)^{\gamma-1}. \tag{8}$$

Equations (5)–(8) are the known Sibuya input. The boundary estimate and its quantum consequence are derived next.

III. ORDINARY-TIME FOURIER BOUND

$$\boxed{|F_t(z)| \leq \frac{2}{|1-z|}\left[\frac{\Gamma(t+\gamma)}{\Gamma(\gamma)\Gamma(t+1)} + \frac{\Gamma(t+1-\gamma)}{\Gamma(1-\gamma)\Gamma(t+1)}\right]} \qquad (t \geq 1,\ |z| = 1,\ z \neq 1). \tag{9}$$

To prove Eq. (9), fix t and put $p_n = a_{t-n} b_n$. Direct cancellation of gamma functions yields

$$\frac{p_{n+1}}{p_n} = \frac{t-n}{t-n-1+\gamma}\frac{n+1-\gamma}{n+1}, \tag{10}$$

and hence

$$\operatorname{sgn}\left(\frac{p_{n+1}}{p_n} - 1\right) = \operatorname{sgn}\big[n+1-\gamma(t+1)\big]. \tag{11}$$

The positive sequence $p_0, \ldots, p_t$ first decreases and then increases, with at most a two-point flat minimum. Its total variation consequently satisfies

$$\sum_{n=1}^{t} |p_n - p_{n-1}| \leq p_0 + p_t. \tag{12}$$

For every complex z, exact discrete summation by parts gives

$$(1-z)F_t(z) = p_0 + \sum_{n=1}^{t}(p_n - p_{n-1})z^n - p_t z^{t+1}. \tag{13}$$

When $|z| = 1$ and $z \neq 1$, the triangle inequality and Eq. (12) prove Eq. (9). In particular, the argument controls the ordinary coefficient sequence and includes all nontrivial roots of unity.

For $t \geq 1$, the standard gamma-ratio inequalities for $0 < \gamma < 1$ give

$$a_t < \frac{t^{\gamma-1}}{\Gamma(\gamma)}, \qquad b_t < \frac{t^{-\gamma}}{\Gamma(1-\gamma)}. \tag{14}$$

Thus Eq. (9) implies

$$|F_t(z)| < \frac{2}{|1-z|}\left[\frac{t^{\gamma-1}}{\Gamma(\gamma)} + \frac{t^{-\gamma}}{\Gamma(1-\gamma)}\right]. \tag{15}$$

Both powers decay for every fixed $0 < \gamma < 1$.

IV. TRACE-NORM CONVERGENCE

Let u_j be the distinct eigenvalues of U, with orthogonal spectral projectors Π_j. Set

$$|\psi_j\rangle = \Pi_j |\psi_R\rangle, \qquad w_j = \|\psi_j\|, \tag{16}$$

and let $J = \{j : w_j > 0\}$ be the occupied set. The pinching onto the distinct eigenspaces of U is

$$\mathcal{D}_U(X) = \sum_j \Pi_j X \Pi_j. \tag{17}$$

The convention $U\rho U^\dagger$ fixes the coherence ratio as $z_{jk} = u_j \overline{u_k}$. Equations (3) and (7) then give the exact block

identity

$$\rho(t) - \mathcal{D}_U(\rho_R) = \sum_{\substack{j,k\in J \\ j\neq k}} F_t(u_j\overline{u_k})|\psi_j\rangle\langle\psi_k|. \tag{18}$$

Because the Schatten trace norm of the rank-one block $|\psi_j\rangle\langle\psi_k|$ is $w_j w_k$, applying Eq. (9) term by term proves

$$\boxed{\|\rho(t) - \mathcal{D}_U(\rho_R)\|_1 \leq 2(a_t + b_t) \sum_{\substack{j,k\in J \\ j\neq k}} \frac{w_j w_k}{|1 - u_j\overline{u_k}|}.} \tag{19}$$

For a coarser form, put $m = |J|$ and, when $m \geq 2$,

$$\delta = \min_{\substack{j,k\in J \\ j\neq k}} |1 - u_j\overline{u_k}| > 0. \tag{20}$$

Since $\sum_j w_j^2 = 1$ and $(\sum_j w_j)^2 - 1 \leq m - 1$, Eqs. (14) and (19) yield

$$\|\rho(t) - \mathcal{D}_U(\rho_R)\|_1 < \frac{2(m-1)}{\delta}\left[\frac{t^{\gamma-1}}{\Gamma(\gamma)} + \frac{t^{-\gamma}}{\Gamma(1-\gamma)}\right]. \tag{21}$$

The right-hand side tends to zero. We have therefore proved

$$\boxed{\lim_{t\to\infty} \|\rho(t) - \mathcal{D}_U(\rho_R)\|_1 = 0} \tag{22}$$

for every fixed finite-dimensional U in the stated model. Exact degeneracies are retained inside one pinched block. If $m = 1$, the difference in Eq. (18) is identically zero and no gap is required.

Equation (21) also gives a conservative clock-time certificate. With $\beta = \min(\gamma, 1-\gamma)$, a sufficient condition for trace-norm error at most ε is

$$t \geq \left[\frac{2(m-1)}{\delta\varepsilon}\left(\frac{1}{\Gamma(\gamma)} + \frac{1}{\Gamma(1-\gamma)}\right)\right]^{1/\beta}. \tag{23}$$

This estimate is not claimed to be sharp, and its divergence as $\delta \to 0$ is essential to its fixed-gap scope.

V. TWO BOUNDARY SINGULARITIES AND PARITY

For fixed $|z| = 1$ with $z \neq 1$, the generating function in Eq. (8) has two distinct algebraic singularities on its circle of convergence, at $u = 1$ and $u = z^{-1}$. Expanding the analytic cofactor at both points and applying the two-point Darboux coefficient estimate [4] gives

$$F_t(z) = (1-z)^{\gamma-1}a_t + (1-z^{-1})^{-\gamma}z^t b_t + O_{\gamma,z}\left(t^{\gamma-2} + t^{-\gamma-1}\right). \tag{24}$$

The branches are those inherited by the binomial series for $|u| < 1$. Subtracting the two displayed singular terms leaves local orders $O((1-u)^{1-\gamma})$ at $u = 1$ and $O((1-zu)^{\gamma})$ at $u = z^{-1}$; their coefficients give the two remainder powers in Eq. (24). The constants can be chosen uniformly for γ in a compact subset of $(0,1)$ and $|1-z| \geq \delta_0 > 0$, but not as the two singularities merge.

Using the gamma-ratio expansions of a_t and b_t gives the leading form

$$F_t(z) = \frac{(1-z)^{\gamma-1}}{\Gamma(\gamma)}t^{\gamma-1} + \frac{(1-z^{-1})^{-\gamma}}{\Gamma(1-\gamma)}z^t t^{-\gamma} + O_{\gamma,z}\left(t^{\gamma-2} + t^{-\gamma-1}\right). \tag{25}$$

The $u = z^{-1}$ contribution decays more slowly when $\gamma < 1/2$, the $u = 1$ contribution decays more slowly when $\gamma > 1/2$, and both are of the same order at $\gamma = 1/2$. Keeping only the $u = 1$ singularity would miss the phase-dependent term relevant to unitary coherences.

The symmetric phase-π point has an exact cancellation. At $\gamma = 1/2$ and $z = -1$,

$$\sum_{t\geq 0} F_t(-1)u^t = (1-u)^{-1/2}(1+u)^{-1/2} = (1-u^2)^{-1/2}, \tag{26}$$

so

$$\boxed{F_{2q+1}(-1) = 0, \qquad F_{2q}(-1) = \frac{\binom{2q}{q}}{4^q} \sim \frac{1}{\sqrt{\pi q}}.} \tag{27}$$

This is a cancellation within convergence, not a resonance or a failure to relax.

Figure 1 evaluates exact coefficients from Eq. (8). Differentiating that generating function gives the recurrence

$$(t+1)F_{t+1}(z) = \big[(1+z)t + \gamma + (1-\gamma)z\big]F_t(z) - ztF_{t-1}(z), \tag{28}$$

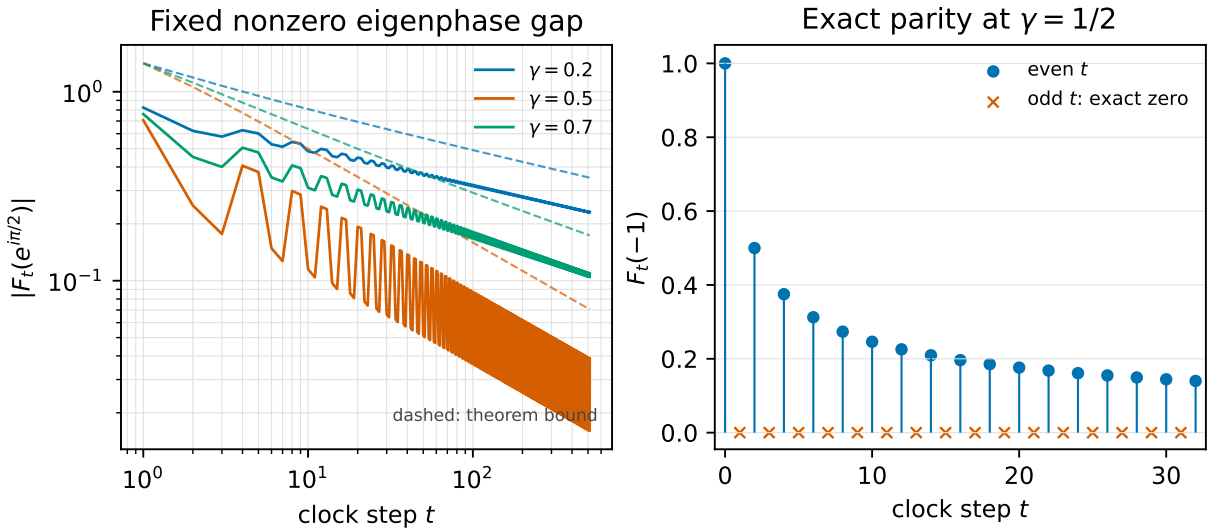


FIG. 1. Critical coherence coefficients. (a) Absolute coefficients at the fixed eigenphase ratio $z = e^{i\pi/2}$ for three values of γ; dashed curves are the exact endpoint-mass bounds in Eq. (9). (b) At $\gamma = 1/2$ and $z = -1$, odd-time coefficients are exactly zero, while even-time coefficients equal $\binom{2q}{q}/4^q$. Values were generated by Eq. (28); the accompanying comma-separated table records every plotted value. The plotting code and figure preparation used the AI assistance disclosed below; the coefficients were checked against an independent 80-digit convolution, the proved envelope, and the exact parity identity.

with $F_0 = 1$ and $F_1 = \gamma + (1-\gamma)z$. The dashed curves use the nonasymptotic right-hand side of Eq. (9); they are not fits. The second panel displays Eq. (27) directly.

VI. PHYSICAL INTERPRETATION AND LIMITATIONS

The limit $\mathcal{D}_U(\rho_R)$ preserves precisely the coherences lying within a degenerate eigenspace of the gate. Every occupied coherence between distinct gate eigenvalues is suppressed, even though the critical renewal time has infinite mean. The two endpoint masses have a direct renewal interpretation: $a_t = P_0(t)$ weights a reset at the observation time, whereas $b_t = P_t(t)$ weights an excursion with no reset after initialization. Their competing algebraic decays determine the slowest fixed-gap coherence scale.

The theorem is finite-dimensional in an essential way: only finitely many occupied nonzero chord gaps enter Eq. (19). It does not provide a bound uniform in a sequence of gates whose occupied gap closes, nor does it exchange the long-time limit with a thermodynamic or infinite-dimensional limit. The endpoint values $\gamma = 0, 1$, hazards with $\alpha \neq 1$, mixed or partial reset rules, time-dependent gates, and dissipative evolution between resets require separate analyses. The norm in Eqs. (19)–(22) is the Schatten trace norm; the conventional trace distance is one half of it.

The numerical calculations accompanying this paper are diagnostics of the closed forms and finite-register block identity. They are not the certificate for the infinite-time result: that certificate is the exact age law, variation estimate, summation-by-parts identity, and finite spectral decomposition above.

VII. CONCLUSION

Critical Sibuya resetting prepares the U-pinched reset state on every fixed finite unitary register. A U-shaped exact age distribution supplies an ordinary-time Fourier bound whose endpoint masses decay for all $0 < \gamma < 1$, and the finite spectral decomposition turns it into an explicit trace-norm estimate. The two distinct unit-circle singularities explain the phase-dependent algebraic rates and the exact half-Sibuya parity effect. A uniform closing-gap crossover and the full classification away from the critical hazard remain open extensions.

DATA AVAILABILITY

No experimental or externally collected data were used. The manuscript source package includes the code and comma-separated data that regenerate Fig. 1, together with independent coefficient and finite-register diagnostic programs. The analytic result is self-contained in the displayed identities and inequalities.

AI-ASSISTED METHODS AND DISCLOSURE

OpenAI Codex, using the GPT-6 Astra and GPT-5.6 Sol model families, was used substantively for literature synthesis, derivations, code generation, diagnostic calculations, an independent model-context replay, figure generation, and manuscript drafting. The workflow fixed the model and claim before review and checked outputs against versioned primary sources, exact algebraic identities, independently implemented recurrences, and a separate replay. Artificial intelligence is not an author. Before any submission, the named author must independently review and refine all AI-assisted content, confirm the tool terms and rights, and assume responsibility for the work.

[1] S. Wald, L. H. Yao, T. Platini, C. Hooley, and F. Carollo, Stochastic resetting in discrete-time quantum dynamics: Steady states and correlations in few-qubit systems, Quantum **9**, 1742 (2025), arXiv:2410.11497 [quant-ph].

[2] T. M. Michelitsch, G. D'Onofrio, F. Polito, and A. P. Riascos, Random walks with stochastic resetting in complex networks: A discrete-time approach, Chaos **35**, 013119 (2025), arXiv:2409.08394 [math.PR].

[3] F. J. Sevilla and A. Valdés-Hernández, Dynamics of closed quantum systems under stochastic resetting, Journal of Physics A: Mathematical and Theoretical **56**, 034001 (2023), arXiv:2209.05643 [quant-ph].

[4] P. Flajolet and R. Sedgewick, *Analytic Combinatorics* (Cambridge University Press, Cambridge, 2009).